Page No.
Date

Q13. $(256)^{0.16} \cdot (256)^{0.9}$

$\Rightarrow (256)^{0.16 + 0.9}$

$\Rightarrow (256)^{\frac{0.25}{100}}$

$\Rightarrow (256)^{\frac{25}{100}}$

$\Rightarrow (256)^{1/4}$

$\Rightarrow (256)^{1/4} = (4)^{\cancel{4} \times \frac{1}{\cancel{4}}}$

$\Rightarrow$ 4 ans

$(a^m \times a^n = a^{m+n})$

4 [2	256
2	128
4 [2	64
2	32
4 [2	16
2	8
4 [2	4
2	2
	1

12) simplify

$$\left(\frac{x^a}{x^{-b}}\right)^{a-b} \cdot \left(\frac{x^b}{x^{-c}}\right)^{b-c} \cdot \left(\frac{x^c}{x^{-a}}\right)^{c-a}$$

sol: $\left(x^{a-(-b)}\right)^{a-b} \cdot \left(x^{b-(-c)}\right)^{b-c} \cdot \left(x^{c-(-a)}\right)^{c-a}$

$\Rightarrow \left(x^{a+b}\right)^{a-b} \cdot \left(x^{b+c}\right)^{b-c} \cdot \left(x^{c+a}\right)^{c-a}$

$\Rightarrow \left(x^{a^2-b^2}\right) \cdot \left(x^{b^2-c^2}\right) \cdot \left(x^{c^2-a^2}\right)$

$\Rightarrow$ (Base same) (power diff.) (powers will add)

$\Rightarrow (x)^{(a^2-b^2)+(b^2-c^2)+(c^2-a^2)}$

$\Rightarrow (x)^0 = 1$

Ans = 1

2 june. 2022.

Page No.
Date

Q.1 if $a = 1 + \sqrt{7}$, find the value of $\frac{-6}{a}$

sol: $\frac{-6}{1+\sqrt{7}} \times \frac{1-\sqrt{7}}{1-\sqrt{7}} = \frac{-6+6\sqrt{7}}{(1)^2-(\sqrt{7})^2}$

$= \frac{-6+6\sqrt{7}}{1-7} = \frac{-6+6\sqrt{7}}{-6} = \frac{\cancel{6}(1+\sqrt{7})}{-\cancel{6}}$

$= \cancel{\sqrt{7}-1}\ 1-\sqrt{7}$

Q.2 if $p = 5 - 2\sqrt{6}$, find : $p^2 + \frac{1}{p^2}$

sol: $\frac{1}{p} = \frac{1}{5-2\sqrt{6}} \times \frac{5+2\sqrt{6}}{5+2\sqrt{6}} = \frac{5+2\sqrt{6}}{(5)^2-(2\sqrt{6})^2}$

$\Rightarrow \frac{5+2\sqrt{6}}{25-24} = \frac{5+2\sqrt{6}}{1} = 5+2\sqrt{6}$

$\cdot\ p^2 + \frac{1}{p^2} \Rightarrow (5-2\sqrt{6})^2 + (5+2\sqrt{6})^2$

$\Rightarrow 25 + 24 - 2.5.\cancel{2\sqrt{6}} + 25 + 24 + 2.5.\cancel{2\sqrt{6}}$

$\Rightarrow 49 + 49$

$\Rightarrow 98$

Q3. if $\sqrt{2} = 1.414$, find : $\sqrt{8} + \sqrt{50} + \sqrt{72} + \sqrt{98}$

sol: $2\sqrt{2} + 5\sqrt{2} + 6\sqrt{2} + 7\sqrt{2}$

$20\sqrt{2}$

$\Rightarrow 20 \times 1.414$

$\Rightarrow 28.28$

Page No.
Date

Q4. Find :

$$\frac{4}{(216)^{-2/3}} + \frac{1}{(256)^{-3/4}} + \frac{2}{(243)^{-1/5}}$$

$$\Rightarrow \frac{4}{(6)^{3\times -2/3}} + \frac{1}{(4)^{4\times -3/4}} +$$

Ans -

$$\Rightarrow \frac{4}{(6)^{3\times -2/3}} + \frac{1}{(4)^{4\times -3/4}} + \frac{2}{(3)^{-1/5\times 5}}$$

$$\Rightarrow \frac{4}{(6)^{-2}} + \frac{1}{(4)^{-3}} + \frac{2}{(3)^{-1}}$$

$\Rightarrow 36\times 4 + (4\times) + 3\times 2$

$\Rightarrow 144 + 64 + 6$

$\Rightarrow 214$ Ans

Answers

Q5. if $n = 9 - 4\sqrt{5}$ find.

a. $n + \frac{1}{n}$ d. $n - \frac{1}{n}$ g. $n + \frac{14}{n}$

b. $n^3 + \frac{1}{n^3}$ e. $n^3 - \frac{1}{n^3}$ h. $n^2 - \frac{1}{n^2}$

c. $n^2 + \frac{1}{n^2}$ f. $n^4 + \frac{1}{n^4}$

Spiral

Page No.
Date

$\left\{ \cdot\, n = 9 - 4\sqrt{5} \quad \cdot\, \frac{1}{n} = 9 + 4\sqrt{5} \right\}$

Answer- (5th Question)

1.) $\frac{1}{n} = \frac{1}{9-4\sqrt{5}} \times \frac{9+4\sqrt{5}}{9+4\sqrt{5}} = \frac{9+4\sqrt{5}}{(9)^2-(4\sqrt{5})^2} = \frac{9+4\sqrt{5}}{81-80} = 9+4\sqrt{5}$

$\Rightarrow 9 - 4\sqrt{5} + 9 + 4\sqrt{5}$

$\Rightarrow 18$ Ans (a.)

$\left(n + \frac{1}{n}\right)$ Ans

2.) $n^3 + \frac{1}{n^3} \Rightarrow (9-4\sqrt{5})^3 + (9+4\sqrt{5})^3$

$\Rightarrow 729 - 320\sqrt{5} - 3(9)^2(4\sqrt{5}) + 3(9)(4\sqrt{5})^2 +$

$729 + 320\sqrt{5} + 3(9)^2(4\sqrt{5}) + 3(9)(4\sqrt{5})^2$

$\Rightarrow 729 - 320\sqrt{5} - 972\sqrt{5} + 2160 + 729 + 972\sqrt{5} + 2160 + 320\sqrt{5}$

$\Rightarrow 729 + 2160 + 729 + 2160$

$\Rightarrow 5778$ Ans (b)

3.) $n^2 + \frac{1}{n^2}$

$\Rightarrow (9-4\sqrt{5})^2 + (9+4\sqrt{5})^2$

$\Rightarrow (9)^2 + (4\sqrt{5})^2 - 2\cdot 9\cdot 4\sqrt{5} + (9)^2 + (4\sqrt{5})^2 + 2\cdot 9\cdot 4\sqrt{5}$

$\Rightarrow 81 + 80 + 81 + 80$

$\Rightarrow 322$ Ans (c.)

4.) $n - \frac{1}{n} \Rightarrow (9-4\sqrt{5}) - (9+4\sqrt{5})$

$\Rightarrow 9 - 4\sqrt{5} - 9 - 4\sqrt{5}$

$\Rightarrow -8\sqrt{5}$ Ans (d.)

5.) $n^3 - \frac{1}{n^3} \Rightarrow (9-4\sqrt{5})^3 - (9+4\sqrt{5})^3$

$\Rightarrow (729 - 320\sqrt{5} - 972\sqrt{5} + 2160) - (729 + 320\sqrt{5} + 2160 + 972\sqrt{5})$

$\Rightarrow 729 - 320\sqrt{5} - 972\sqrt{5} + 2160 - 729 - 320\sqrt{5} - 2160 - 972\sqrt{5}$

$\Rightarrow -640\sqrt{5} - 1944\sqrt{5}$

$\Rightarrow -2584\sqrt{5}$ Ans (e.)

Spiral

Page No.
Date

5.) $x^4 + \frac{1}{x^4} \Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = (322)^2$

$\Rightarrow \left(x^4 + \frac{1}{x^4}\right) + 2 \cdot x^2 \cdot \frac{1}{x^2} = 103684$

$\Rightarrow x^4 + \frac{1}{x^4} = 103684 - 2$

$\Rightarrow x^4 + \frac{1}{x^4} = 103682$ (f.)

Q6. $a = \frac{\sqrt{7} - \sqrt{6}}{\sqrt{7} + \sqrt{6}}$, $b = \frac{\sqrt{7} + \sqrt{6}}{\sqrt{7} - \sqrt{6}}$, find $a^2 + b^2$

sol: $a = \frac{\sqrt{7} - \sqrt{6}}{\sqrt{7} + \sqrt{6}} \times \frac{\sqrt{7} - \sqrt{6}}{\sqrt{7} - \sqrt{6}} = (\sqrt{7} - \sqrt{6})^2 = 7 + 6 - 2\sqrt{42}$
$= 13 - 2\sqrt{42}$

$b = 13 + 2\sqrt{42}$

$\Rightarrow (13 + 2\sqrt{42})^2 + (13 - 2\sqrt{42})^2 + (13 - 2\sqrt{42})(13 + 2\sqrt{42})$

$169 + 168 + 52\sqrt{42} + 169 + 168 - 52\sqrt{42} + 169 - 168$

$\Rightarrow 169 + 169 + 168 + 169$

$\Rightarrow 675$

Q7. simplify:

$$\frac{1}{1 + x^{b-a} + x^{c-a}} + \frac{1}{1 + x^{a-b} + x^{c-b}} + \frac{1}{1 + x^{a-c} + x^{b-c}}$$

Q8. show that:

$$\frac{1}{3\sqrt{2} + \sqrt{8}} - \frac{1}{\sqrt{8} - \sqrt{7}} + \frac{1}{\sqrt{7} - \sqrt{6}} - \frac{1}{\sqrt{6} - \sqrt{5}} + \frac{1}{\sqrt{5} - 2} = 5$$

Page No.
Date

Q9. Evalute

$$\frac{1}{\sqrt{2} + 1} + \frac{1}{\sqrt{3} + \sqrt{2}} + \frac{1}{\sqrt{4} + \sqrt{3}} \cdots\cdots + \frac{1}{\sqrt{9} + \sqrt{8}}$$

~~Q7~~ Answer -

7.)

Ans. $\frac{1}{\frac{1}{1} + \frac{x^a}{x^a} + \frac{x^c}{x^a}} + \frac{1}{\frac{1}{1} + \frac{x^a}{x^b} + \frac{x^c}{x^b}} + \frac{1}{1 + \frac{x^a}{x^c} + \frac{x^b}{x^c}}$

$\Rightarrow \frac{1}{\frac{x^a + x^b + x^c}{x^a}} + \frac{1}{\frac{x^a + x^b + x^c}{x^b}} + \frac{1}{\frac{x^a + x^b + x^c}{x^c}}$

$\Rightarrow \frac{x^a}{x^a + x^b + x^c} + \frac{x^b}{x^a + x^b + x^c} + \frac{x^c}{x^a + x^b + x^c}$

$\Rightarrow \frac{x^a + x^b + x^c}{x^a + x^b + x^c} = \frac{1}{1} = 1$ Ans

(Ans)

Ans-8

$$\frac{1}{3-\sqrt{8}} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2} = 5$$

- $\frac{1}{3-\sqrt{8}} \times \frac{3+\sqrt{8}}{3+\sqrt{8}} = \frac{3+\sqrt{8}}{(3)^2-(\sqrt{8})^2} = 3+\sqrt{8}$

- $\frac{1}{\sqrt{8}-\sqrt{7}} \times \frac{\sqrt{8}+\sqrt{7}}{\sqrt{8}+\sqrt{7}} = \frac{\sqrt{8}+\sqrt{7}}{(\sqrt{8})^2-(\sqrt{7})^2} = \sqrt{8}+\sqrt{7}$

- $\frac{1}{\sqrt{7}-\sqrt{6}} \times \frac{\sqrt{7}+\sqrt{6}}{\sqrt{7}+\sqrt{6}} = \frac{\sqrt{7}+\sqrt{6}}{(\sqrt{7})^2-(\sqrt{6})^2} = \sqrt{7}+\sqrt{6}$

- $\frac{1}{\sqrt{6}-\sqrt{5}} \times \frac{\sqrt{6}+\sqrt{5}}{\sqrt{6}+\sqrt{5}} = \frac{\sqrt{6}+\sqrt{5}}{(\sqrt{6})^2-(\sqrt{5})^2} = \sqrt{6}+\sqrt{5}$

- $\frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} = \frac{\sqrt{5}+2}{(\sqrt{5})^2-(2)^2} = \sqrt{5}+2$

$(3-\sqrt{8}) - (\sqrt{8}+\sqrt{7}) + (\sqrt{7}+\sqrt{6}) - (\sqrt{6}+\sqrt{5}) + (\sqrt{5}+2)$

$3 - \cancel{\sqrt{8}} + \cancel{\sqrt{8}} - \cancel{\sqrt{7}} + \cancel{\sqrt{7}} + \cancel{\sqrt{6}} - \cancel{\sqrt{6}} + \cancel{\sqrt{5}} + 2 - \cancel{\sqrt{5}}$

$\Rightarrow 3 + 2 = 5$

$5 = 5$

Ans-9 Evalute

$$\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{6}} +$$

$$\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} + \frac{1}{\sqrt{5}+\sqrt{4}} + \frac{1}{\sqrt{6}+\sqrt{5}} + \frac{1}{\sqrt{7}+\sqrt{6}} +$$

$$\frac{1}{\sqrt{8}+\sqrt{7}} + \frac{1}{\sqrt{9}+\sqrt{8}}.$$

sol:

- $\frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = \frac{\sqrt{2}-1}{(\sqrt{2})^2-(1)^2} = \sqrt{2}-1$

- $\frac{1}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{\sqrt{3}-\sqrt{2}}{(\sqrt{3})^2-(\sqrt{2})^2} = \sqrt{3}-\sqrt{2}$

- $\frac{1}{\sqrt{4}+\sqrt{3}} \times \frac{\sqrt{4}-\sqrt{3}}{\sqrt{4}-\sqrt{3}} = \frac{\sqrt{4}-\sqrt{3}}{(\sqrt{4})^2-(\sqrt{3})^2} = \sqrt{4}-\sqrt{3}$

- $\frac{1}{\sqrt{5}+\sqrt{4}} \times \frac{\sqrt{5}-\sqrt{4}}{\sqrt{5}-\sqrt{4}} = \frac{\sqrt{5}-\sqrt{4}}{(\sqrt{5})^2-(\sqrt{4})^2} = \sqrt{5}-\sqrt{4}$

- $\frac{1}{\sqrt{6}+\sqrt{5}} \times \frac{\sqrt{6}-\sqrt{5}}{(\sqrt{6}-\sqrt{5})} = \frac{\sqrt{6}-\sqrt{5}}{(\sqrt{6})^2-(\sqrt{5})^2} = \sqrt{6}-\sqrt{5}$

- $\frac{1}{\sqrt{7}+\sqrt{6}} \times \frac{\sqrt{7}-\sqrt{6}}{\sqrt{7}-\sqrt{6}} = \frac{\sqrt{7}-\sqrt{6}}{(\sqrt{7})^2-(\sqrt{6})^2} = \sqrt{7}-\sqrt{6}$

- $\frac{1}{\sqrt{8}+\sqrt{7}} \times \frac{\sqrt{8}-\sqrt{7}}{\sqrt{8}-\sqrt{7}} = \frac{\sqrt{8}-\sqrt{7}}{(\sqrt{8})^2-(\sqrt{7})^2} = \sqrt{8}-\sqrt{7}$

$\cdot \frac{1}{\sqrt{9}+\sqrt{8}} \times \frac{\sqrt{9}-\sqrt{8}}{\sqrt{9}-\sqrt{8}} = \frac{\sqrt{9}-\sqrt{8}}{(\sqrt{9})^2-(\sqrt{8})^2} = \sqrt{9}-\sqrt{8}$

Adding all -

$\cdot$ $(\sqrt{2}-1)+(\sqrt{3}-\sqrt{2})+(\sqrt{4}-\sqrt{3})+(\sqrt{5}-\sqrt{4})+(\sqrt{6}-\sqrt{5})+(\sqrt{7}-\sqrt{6})+(\sqrt{8}-\sqrt{7})+(\sqrt{9}-\sqrt{8})$

$\cdot$ $\sqrt{2}-1+\sqrt{3}-\sqrt{2}+\sqrt{4}-\sqrt{3}+\sqrt{5}-\sqrt{4}+\sqrt{6}-\sqrt{5}+\sqrt{7}-\sqrt{6}+\sqrt{8}-\sqrt{7}+\sqrt{9}-\sqrt{8}$

$\Rightarrow -1+\sqrt{9}$

$\Rightarrow \sqrt{9}-1$ Ans

Q 5.

7.) $n+\frac{14}{n}$

$\Rightarrow (9-4\sqrt{5})+\left(\frac{14}{9+4\sqrt{5}}\right)$

$\Rightarrow 9-4\sqrt{5}+\frac{14}{9-4\sqrt{5}} \times \frac{9+4\sqrt{5}}{9+4\sqrt{5}} \Rightarrow \frac{126+56\sqrt{5}}{(9)^2-(4\sqrt{5})^2}$

$= 126+56\sqrt{5}$

$\Rightarrow 9-4\sqrt{5}+126+56\sqrt{5}$

$\Rightarrow 135+52\sqrt{5}$

8.) $n^2-\frac{1}{n^2}$

$\Rightarrow (9-4\sqrt{5})^2-(9+4\sqrt{5})^2$

$\Rightarrow (81+80-2.9.4\sqrt{5})-(81+80+2.9.4\sqrt{5})$

$\Rightarrow \cancel{81}+\cancel{80}-2.9.4\sqrt{5}-\cancel{81}-\cancel{80}-2.9.4\sqrt{5}$

$\Rightarrow -72\sqrt{5}-72\sqrt{5} = -144\sqrt{5}$

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Ques- 10

$[256]^{-\left(4^{-\frac{3}{2}}\right)}$

sol: $[256]^{-\left(2^{2\times-\frac{3}{2}}\right)}$

$\Rightarrow (256)^{-\left(2^{-3}\right)}$

$= (256)^{-\left(\frac{1}{2^3}\right)}$

$= (256)^{-\left(\frac{1}{8}\right)}$

$= \left(\frac{1}{256}\right)^{\frac{1}{8}}$

$= \left(\frac{1}{2}\right)^{8\times\frac{1}{8}}$

$= \frac{1}{2}^{1} = \frac{1}{2}$ Ans

Q. $(256)^{-\left(4^{-\frac{3}{2}}\right)}$

$\Rightarrow (256)^{-\left(2^{2\times-\frac{3}{2}}\right)}$

$= (256)^{-\left(2^{-3}\right)}$

$= (256)^{-\left(\frac{1}{8}\right)}$

$= \frac{1}{(256)^{\frac{1}{8}}}$

$= \frac{1}{2^{8\times\frac{1}{8}}}$

$= \frac{1}{2}$ Ans

$(256)^{-\left(4^{-\frac{3}{2}}\right)}$

$(256)^{-\left(\frac{1}{4^{3/2}}\right)}$

$(256)^{\frac{3}{8}}$

$(256)^{8(2)^{3\times\frac{1}{8}}}$

(256)

$(256)^{(4)^{-\frac{3}{2}}}$

$\Rightarrow (256)^{-\left(\frac{1}{4^{3/2}}\right)}$

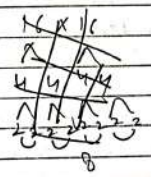
$\Rightarrow (256)^{-\left(\frac{1}{2^{2\times-\frac{3}{2}}}\right)}$

$\Rightarrow (256)^{-\frac{1}{2^3}}$

$\Rightarrow (256)^{-\frac{1}{8}}$

$\Rightarrow \left(\frac{1}{256}\right)^{1/8}$

$\Rightarrow \frac{1}{2}^{8\times1/8} \Rightarrow$



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